ISE 555: Unit 5 Assignment

Fall 2026

For this assignment, all problems, including the writing assignment (Problem 5), are mandatory to receive full credit. A PDF file containing all your solutions as well as the writing assignment must be submitted on Brightspace by **4:00 pm on Wednesday, October 21st.** Make sure you have access to your submission during class on the day that the assignment is due. This will be helpful when presenting your solutions to the class in person.
Please type your solutions if possible. Suggest to use LaTeX (available https://overleaf.com)

1) Solve the following problem using the Big-M Simplex Method. Show your tableau at each iteration.

$$\begin{aligned} \text{minimize} \quad & z = -4x_1 - 2x_2 \\ \text{subject to} \quad & 3x_1 - 2x_2 \geq 4 \\ & x_1 - 2x_2 = 2 \\ & x_1, x_2 \geq 0 \end{aligned}$$

2) Write the following linear programming problem in three forms: standard form, canonical primal form, and dual form.

$$\begin{aligned} \text{maximize} \quad & z = 2x_1 - 3x_2 \\ \text{subject to} \quad & x_1 + x_2 \geq 3 \\ & 3x_1 + x_2 \leq 6 \\ & x_1, x_2 \geq 0 \end{aligned}$$

3) Find the dual form of the following linear program.

$$\begin{aligned} \text{minimize} \quad & z = -2x_1 + 4x_2 - 3x_3 \\ \text{subject to} \quad & 9x_1 - 2x_2 - 8x_3 = 5 \\ & 3x_1 + 3x_2 + 3x_3 = 7 \\ & 7x_1 - 5x_2 + 2x_3 = 9 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

4) Solve the following linear program using the regular simplex method. Then write the dual form of the problem and use complementary slackness to determine the dual solution.

$$\begin{aligned} \text{minimize} \quad & z = -x_1 + 2x_2 - x_3 \\ \text{subject to} \quad & x_1 + 2x_2 + x_3 \leq 12 \\ & 2x_1 + x_2 - x_3 \leq 6 \\ & -x_1 + 3x_2 \leq 9 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

5) WRITING ASSIGNMENT: In a previous assignment you had to write about using optimization to determine how to choose which site of a company to close. Write a new paragraph based on your knowledge of dual problems (how a dual can be used to help figure out which to close).